

Blockchain-Powered Individual Carbon Footprint Tracking: A Daily Transportation Perspective

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Abstract—Due to climate change, there is an urgent need for tools that help individuals understand and reduce their carbon emissions. This paper has proposed a blockchain based system which allows the user to track their own carbon footprint and the carbon emitted from daily transportation activities. There is smart contract in the system which calculates the daily carbon emission when it receives the required information like travel distance, type of transport and fuel usage, and the use of smart contract ensures data security, immutability and transparency while calculating the daily emissions. This architecture also supports future improvements such as integration with carbon credit frameworks and incentive mechanisms. By combining blockchain technology with emission analysis, the system empowers individuals to take responsibility for their carbon footprint and make environmentally conscious decisions.

Index Terms—Blockchain, Carbon Footprint, Transportation, Sustainability, Smart Contracts, Decentralized Ledger

I. INTRODUCTION

Transport produces a large amount of pollution that causes global warming and is one of the major causes of global greenhouse gas emissions. Due to growing population there is a significant growth in transportation activities like going to work, traveling from one city to another and goods transportation which all contribute to an individual's carbon footprint and as we know we can't control these activities but we can track the carbon emissions, but currently the tools to track personal carbon emissions are limited.

Most of the available solutions are controlled and managed by a central authority, meaning a single organization, which lacks transparency, raises privacy concerns and makes it harder for any individual or user to trust the emissions calculated.

So, to solve these challenges and to establish trust, we will use blockchain technology to solve these issues because in blockchain networks there is no central authority and no single organization has control over data, so there is transparency. These features make blockchain suitable for work where we have to keep records safe.

II. LITERATURE REVIEW

Most of the existing carbon footprint tracking systems are relying on centralized platforms where a single authority does all the process like calculations and data management, so any individual user can't trust the data or emission calculation done by these authorities, and that's where our system resolves this issue by adding trust and transparency in the calculations and every individual can track the emissions.

Sometimes to improve the estimation and accuracy, AI is used, but still the system remains centralized, so the problems like data integrity and accountability will be there. These issues lead to the development of a decentralized system so the user can have trust and belief in personal carbon tracking systems.

After reviewing the literature, it is seen that many researchers are seeing blockchain as one of the solutions for carbon tracking. The main idea is straightforward: once data is recorded on a blockchain, it becomes difficult to modify, and control is distributed rather than centralized. For instance, Cruz et al. (2020) proposed a Solidity-based smart contract system to trace carbon footprints in supply chains [1]. In this work the researchers here have showed how blockchain can support immutable record keeping. Similarly, Saraji and Borowczak (2021) designed a blockchain-based ecosystem for carbon credit trading where smart contracts automate the transactions in decentralized markets [2]. All these studies shows why blockchain is required but still they are mostly focused on large organizations and carbon markets rather than the individuals so our main goal is to develop one for the individuals.

There are some works which combine blockchain with Internet of Things (IoT) technologies, like Mofatteh et al. (2024) introduced a framework integrating IoT devices for real-time data collection with blockchain for secure reporting in supply chains [4]. Earlier, Kouhizadeh and Sarkis (2018) discussed how blockchain practices could enhance greener

supply chains through improved transparency and traceability [7]. But these contributions only focus on enterprises rather than individuals.

There are some researchers who have studied the role of blockchain in sustainability. Saberi et al. (2019) analyzed how decentralized technologies may improve environmental accountability in supply chain management [8]. Howson (2019) examined blockchain within climate governance and transparent carbon accounting [9]. While these works explain the foundation of blockchain or the need of blockchain in sustainability, they explain nothing about real-time personal emission tracking.

Now in the carbon trading systems, Khaqqi et al. (2018) proposed a reputation-based mechanism for blockchain-supported emission trading platforms to enhance market reliability [10]. Fu et al. (2018) applied blockchain-based emission trading in the fashion manufacturing industry [11]. Pan et al. (2019) demonstrated how blockchain improves transparency and efficiency in carbon markets across industries [12]. But here also the problem is same, they majorly focus on large organisations rather than focusing on the daily carbon emissions of individual users, so the problem for individual users is still there.

After studying and reviewing the literatures, it has become clear that most of the blockchain-based carbon tracking solutions mainly focus on supply chains, industries, or large-scale policy concerns such as energy consumption in Bitcoin mining (Jiang et al., 2021) [3]. Individual users are rarely the primary focus. For example, Li et al. (2018) proposed a blockchain-based cloud manufacturing platform operating as a distributed peer-to-peer system [13], but here also the main objective was industrial collaboration rather than personal carbon monitoring, so here also there is no focus on individual users.

Now, in the energy sector also, observation is there. Andoni et al. (2019) conducted a systematic review of blockchain applications and identified opportunities in emission tracking and carbon credit systems [14]. Here also, the focus was organizational. Sund et al. (2020) demonstrated blockchain-based event processing in a case study at IKEA [15], again focusing on enterprise-level operations rather than individual tracking.

New studies also follow the same direction. The GETS framework proposed by Jiang et al. (2024) supports greenhouse gas disclosures but it manages industrial data [6]. Pekel and Yayla (2024) introduced a tokenized carbon footprint system, yet transportation-specific modeling remained limited [5]. Karamchandani et al. (2020) studied enterprise blockchain adoption in the Indian service industry [16], still focusing on operational improvements rather than personal or individual tracking. Mirabelli and Solina (2020) explored blockchain applications in agricultural supply chains [17], focusing on product-level traceability instead of individual emissions.

So from these points it has become clear that all the previous research works mainly focus on organization level rather than individual carbon footprint tracking and that is what leads to

our research and forms the basis for the comparison presented in Table I.

TABLE I: Comparison of Blockchain-Based Carbon Tracking Systems

Work	Domain	Indiv.	IoT	Limitation
Cruz (2020)	Supply Chain	No	No	Logistics-focused implementation
Saraji (2021)	Carbon Markets	No	No	Market-oriented framework
Pekel (2024)	Tokenized System	Limited	No	Lacks real-time transport tracking
GETS (2024)	Industrial Emissions	No	Yes	Organization-level scope only
Li (2018)	Manufacturing	No	No	Industrial production focus
Mirabelli (2020)	Agriculture	No	No	Product traceability emphasis
BCFT (This Work)	Individual Transport	Yes	Yes	Real-time individual tracking

Table I demonstrates that existing blockchain-based carbon tracking solutions predominantly target organizational or industrial applications. They generally do not support real-time, individual-level transportation monitoring.

From both the literature review and comparative analysis, a clear research gap emerges: there is limited work on building decentralized, user-focused systems capable of tracking personal transportation emissions in real time. Most current solutions operate at the supply chain or carbon trading market level, with minimal emphasis on individual users. This gap directly motivated the development of the proposed Blockchain-based Carbon Footprint Tracking (BCFT) framework.

III. PROPOSED SYSTEM: BLOCKCHAIN BASED CARBON FOOTPRINT TRACKER (BCFT) FRAMEWORK

This paper proposes a Blockchain based Carbon Footprint Tracking Framework. This framework is designed to track personal Carbon Footprints due to transportation instead of tracking large industrial based Emission, main aim is to focus on carbon emissions due to personal choice of transport and commute. The application keeps the track of the user's mode of transport i.e. Road, Air, Water etc., type of vehicle (personal or commercial) and if it is possible to connect the application to the vehicle (via GPS or OBD-2 device) to automatically measure the distance travelled or to put the distance travelled manually and fuel type.

All this data goes into the smart contract where emission calculation is done and stored in the blockchain ledger.

When the user submits the new information it is immediately validated by the smart contract and then it calculates the emissions and stores the results. Also, the data is stored in a unique address which is linked to that particular user who submitted that data, hence data can be stored effectively and can be easily accessed as every user has a unique address which can be used to get all the data of that user.

Hence, we can say that this is a transparent, scalable and personal carbon emission tracking system that any individual

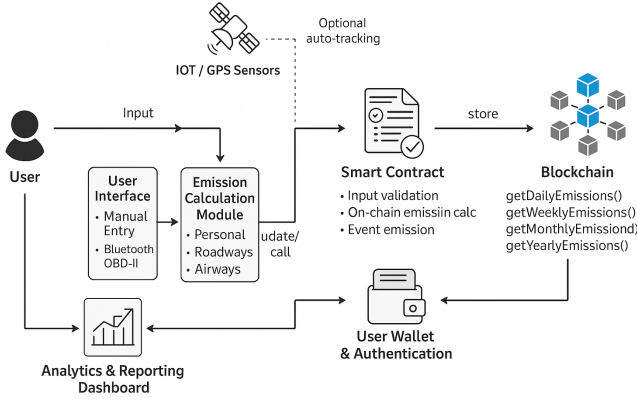


Fig. 1: Basic Architecture of the BCFT Framework

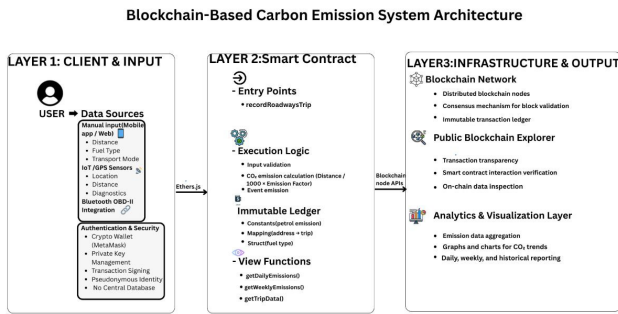


Fig. 2: Detailed Architecture of the BCFT Framework

can use to keep track of their emissions rather than only being used by big enterprises only and only for the complex uses cases.

Also, it is also possible to use this application data to compare the emission over daily, weekly, monthly and yearly for further research and improvement.

A. System Overview

This BCFT framework uses Blockchain Technology to calculate and store the carbon emissions as it provides transparency and decentralization.

B. Architectural Illustration

Figure 1 shows the high-level system architecture, while Figure 2 illustrates detailed interactions between components. These diagrams demonstrate how transportation data is captured, processed via smart contracts, and securely recorded on the blockchain ledger.

C. Core Components

Based on the proposed architecture, the BCFT framework consists of the following elements:

1) Blockchain Network: Decentralized, immutable storage of calculated emission records, once stored, data cannot be changed.

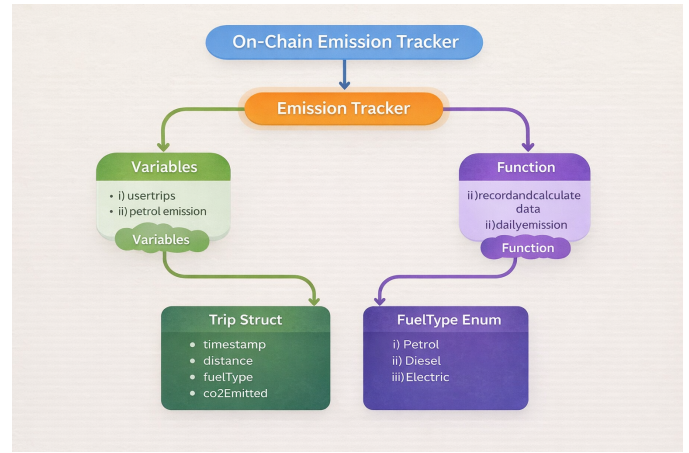


Fig. 3: On-Chain Emission Tracker Smart Contract Architecture

2) Smart Contracts: Receive the user data, verify the correctness of data, and calculate the final carbon emissions based on preconfigured factors and transmits the data to the blockchain.

D. Key Components

1) User Registration and Authentication: Users are identified through cryptographic wallet addresses (e.g., MetaMask). Authentication is handled via blockchain-based digital signatures rather than traditional login credentials.

2) No Central Database: The system eliminates centralized storage, thereby removing single points of failure. User identities remain pseudonymous and are secured by private keys.

3) Transportation Data Collection:

Either users manually input the data such as distance or fuel type or if possible the application can be connected to the bluetooth, OBD-2 device to automatically collect the data.

c) Blockchain Integration: Collected data is transmitted to the blockchain via smart contract functions.

4) Carbon Footprint Calculation:

a) Emission Factors: Predefined constants (e.g., PETROL_EMISSION_gPerKm) are stored within the smart contract. These values are derived from recognized environmental datasets such as the European Environment Agency and Our World in Data.

b) Carbon Emission Calculation:

For personal vehicles, well defined functions calculate emissions based on type of fuel and distance travelled.

For public transport, fixed emission factors are used (e.g., railway emission rates).

5) Smart Contract Verification:

Input Validation: Well defined functions are used such as "recordRoadwaysTrip()" use "require()" statements to reject invalid inputs (e.g., undefined transport categories).

Transparent Logic: All emission calculations are executed on-chain, ensuring auditability and consistency.

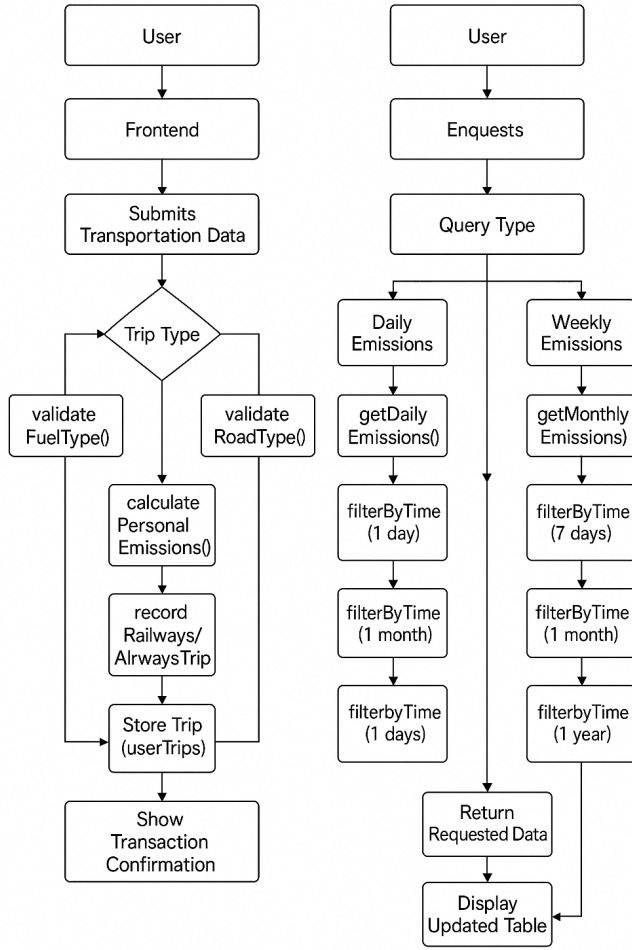


Fig. 4: Workflow of the BCFT Framework

6) Immutable Records: Trips are stored in a "userTrips" mapping linked to the user's wallet address. Public view functions such as "getTripData()" allow retrieval of historical records.

E. Workflow and Data Processing

The operational workflow of the BCFT system is illustrated in Figure 4 and proceeds as follows:

Step 1: Data Collection A user completes a trip and provides input such as Distance and Fuel Type.

Step 2: Transaction Initiation The frontend application (e.g., JavaScript with Ethers.js) invokes the smart contract function: recordPersonalVehicleTrip(uint8 fuelType, uint256 distance).

The transaction is digitally signed via the user's wallet.

Step 3: On-Chain Execution

Validation: The contract verifies input constraints (e.g., fuelType != 3).

Calculation: CO2 emissions are computed using predefined emission factors.

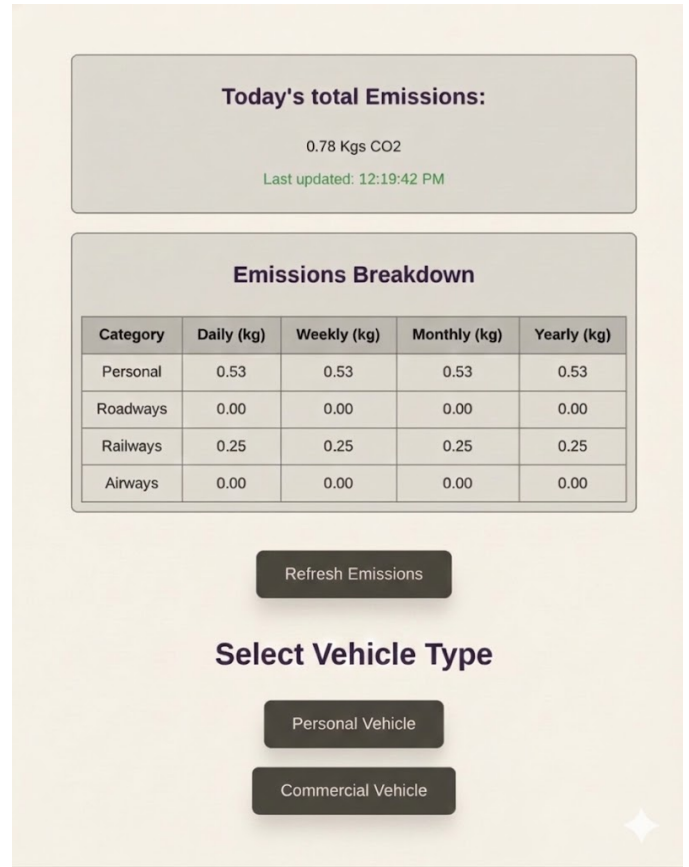


Fig. 5: Dashboard and Emissions Overview Interface

Storage: A "Trip" struct is created and appended to the users address data.

Event Emission: A "TripEnded" event logs transaction details.

Step 4: Blockchain Consensus The transaction is validated by network nodes (e.g., Ethereum validators) and permanently recorded on the blockchain.

Step 5: Data Retrieval User can get their emission information.

F. Application Interface

The BCFT framework includes a decentralized application (DApp) interface that enables users to:

Connect their blockchain wallet, Submit transportation data, View emission summaries, Retrieve historical trip records.

The frontend interacts with the smart contract via Web3 libraries (e.g., Ethers.js), establishing communication between users and the blockchain network.

1) User Interface Functionalities: a) Dashboard and Emissions Overview:

The dashboard displays total CO2 emissions along with a categorized breakdown (Personal, Roadways, Railways, Airways) across multiple timeframes: daily, weekly, monthly, and yearly, as shown in Fig 5. Users can refresh data dynamically and choose between logging a Personal or Commercial vehicle trip.

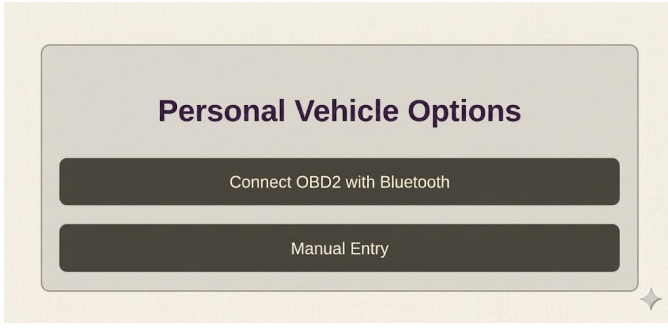


Fig. 6: Personal Vehicle Selection Interface

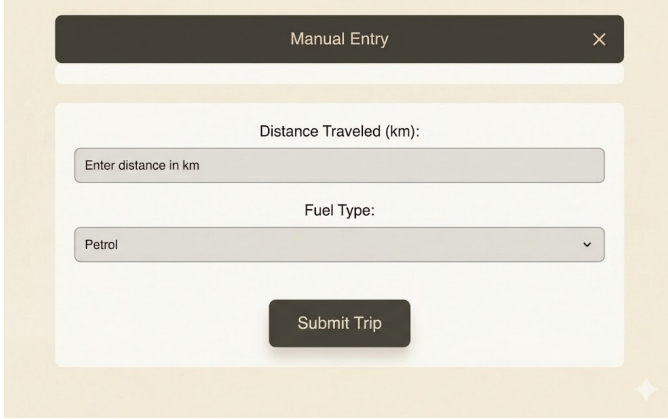


Fig. 7: Manual Trip Entry for Personal Vehicles

b) Personal Vehicle Options: When selecting *Personal Vehicle*, users may connect their vehicle via Bluetooth using an OBD2 device for automatic data capture or manually enter trip details such as fuel type and distance, as shown in Fig. 6.

c) Manual Entry for Personal Vehicles: If manual entry is selected, users input the distance traveled and select the corresponding fuel type before submitting the trip, as shown in Fig. 7. Upon submission, the smart contract updates the user's emission records accordingly.

d) Commercial Vehicle Type Selection: When selecting *Commercial Vehicle*, users are prompted to choose the transportation mode (Railways, Roadways, or Airways), ensuring correct categorization of emissions, as shown in Fig. 8.

e) Commercial Roadways Trip Entry: If *Roadways* is selected, users enter the distance traveled and choose the vehicle type (Taxi, Bus, Auto), as shown in Fig. 9.

f) Commercial Railways/Airways Trip Entry: For Railways or Airways, users simply enter the travel distance and submit the trip, as shown in Fig. 10. The system automatically calculates emissions using predefined emission factors.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Gas Consumption Analysis for BCFT Operations

To test the efficiency, security, and performance of the Blockchain-Based Carbon Footprint Tracking (BCFT) system,

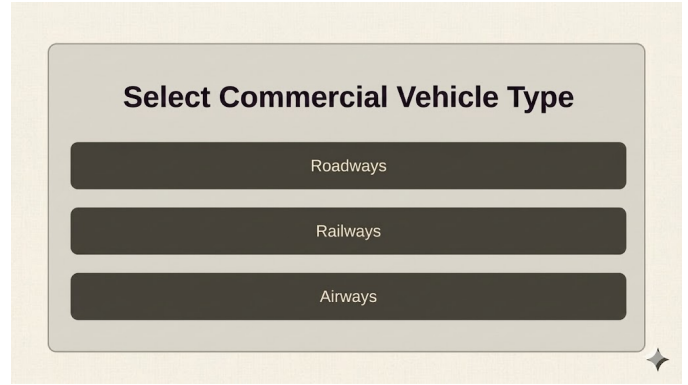


Fig. 8: Commercial Vehicle Mode Selection

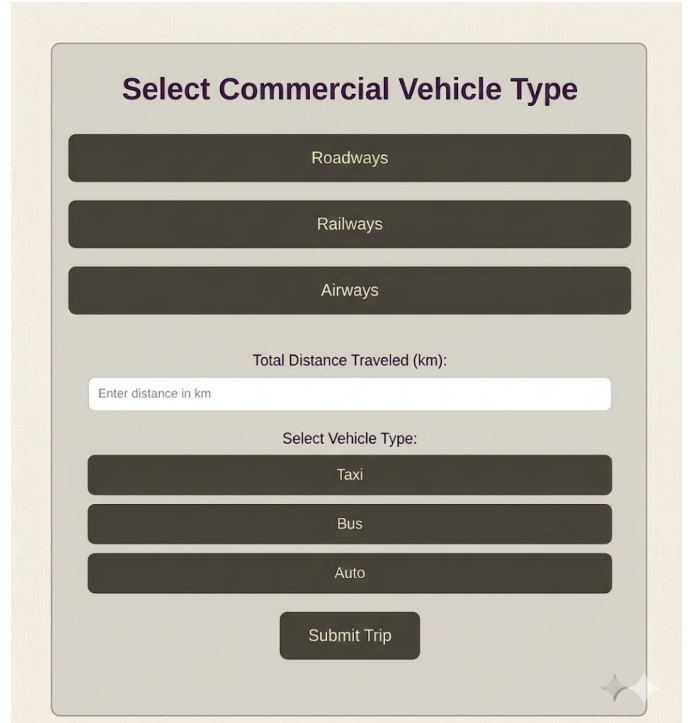


Fig. 9: Commercial Roadways Trip Entry

a complete experimental setup was created. This setup was made to mimic real-world conditions, allowing us to measure performance in a controlled manner.

This configuration allowed us to test: How much gas is used?, The processing speed?, The robustness of the security?

B. Experimental configuration

1) *Development and runtime environment:* Hardware Specifications: CPU: Intel Core i7, RAM: 32 GB, Storage: 1 TB SSD, Operating System: Ubuntu.

Software Stack: Smart Contracts: Solidity, Frontend: HTML, CSS, JavaScript, Web3 Integration: Ethers.js, Development Framework: Hardhat (used for compilation, deployment, and testing locally).

Fig. 10: Commercial Railways/Airways Trip Entry

2) *Blockchain Network Configuration:* The BCFT framework is tested in local and public blockchain environments to compare how it works under controlled conditions with how it behaves in real-world situations.

Local Simulation (Hardhat Network) It is used to test its functionality, determine the amount of gas it consumes, and ensure its security. The gas limit per block is 30,000,000, the same as the Ethereum main network, and it follows the same rules. Mining Modes: Automatic mining to test its proper functioning, and interval mining (12 to 15 seconds) to check for delays.

Public Testnet (Sepolia / Goerli): Used to test transaction confirmation speed, network performance, and congestion. The experimental results in Tables IV and VI were created using specific tools designed to test blockchain systems.

Gas Consumption Analysis: Hardhat Gas Reporter is used to track how much gas is used by each function, Costs are first calculated in ETH then changed to USD. A comparison was made between an optimized version that uses enumerations, events, and constants and a simple version that uses lots of basic logic and storage. The optimized design uses less storage and reduces work from repetitive actions.

Security Auditing: Slither was used for static analysis to find the problems like reentry problems and hidden variables and MythX was used for a more detail analysis of the code meaning and how it runs step by step.

C. Gas Optimization Techniques and Performance Impact

* Zero gas is used for external view call but internal still consumes gas.
*It still uses external call only, internal contract calls still consume gas.

D. Comparative Gas Consumption by Operation Type

*Cached off-chain query.

E. Security and Stress Testing

We also run fuzz testing with the help of **Echidna** to verify smart contract invariants and make sure that the system works correctly in random conditions.

TABLE II: Gas Optimization Techniques and Performance Impact

Technique	Reduction	Benefit
Const Emission Factors	15–20%	Compile-time constants reduce repeated storage reads.
Pure Functions	8–12%	No storage overhead (e.g., <code>calcPersonalEmissions()</code>).
Enum Optimization	5–10%	1-byte enums significantly reduce storage compared to strings.
Event-Based Logging	25–30%	Storage writes minimized via <code>TripEnded</code> events.
View Queries	100%*	Zero gas for external calls (e.g., <code>getDailyEmissions()</code>).
Loop-Free Logic	10–15%	Direct computation avoids costly iterative operations.

TABLE III: Comparative Gas Consumption by Operation

Operation	Base Gas	Optimized Gas	Reduction	Freq.
Personal Vehicle	95k–105k	65k–85k	~25%	High
Commercial Vehicle	85k–95k	60k–80k	~22%	Medium
Railways/Airways	65k–75k	45k–65k	~20%	Low
Emission Query	25k–35k	0*	~100%	Very High

For Load Testing special scripts is used with Mocha and Chai to mimic multiple user using the same system. More than 1,000 trip transactions were executed to test scalability, loop limits, and storage use.

F. Limitations of Experimental Setup

While the setup is used to check the real world situations, there are still some limitations:

Data Source: Emission data was either given by individual people or created by simulated methods. We did not connect directly to car diagnostic systems or mileage tracking tool. **Verification:** Emission calculations are based on fixed numbers from standard guidelines (EPA, DEFRA), we did not use live monitoring and real-time data.

G. Smart Contract Vulnerability Assessment

TABLE IV: Smart Contract Vulnerability Assessment

Test Case	Tool	Result	CVSS Score
Reentrancy Attacks	Slither / MythX	Pass	N/A
Integer Overflow	Echidna	Pass	0.0
Access Control	Manual Testing	Pass	0.0
Gas Limit Exhaustion	Hardhat	Partial Pass*	5.3 (Medium)
Event Consistency	Log Custom Scripts	Pass	0.0

* Performance degradation was observed over 850+ trips in a single block.

H. Performance Discussion

The result shows that BCFT consumes much less gas compared to the standard version. This efficiency is important because transactions are frequent and it helps to reduce the cost.

By using the `view` functions, users can view emission without paying, which is important for data analysis and making reports.

Furthermore, the way the system is optimized helps reduce the computational workload and energy consumption. By reducing the amount of data and removing extra steps, the system saves extra gas and becomes environmentally friendly to fulfill the environment goal of the BCFT framework.

I. Key Security Implementations

The BCFT smart contract uses several checks to ensure proper functioning and to prevent harmful input. Range checks using `require(personalFuelType <= 3)` ensure that the fuel type is selected from the allowed options in a predefined list. Type safety ensures that when inputs are restricted to allowed options, the possibility of incorrect or unexpected values is removed, keeping the system in a safe state. In addition, Solidity 0.8.x provides built-in arithmetic safety that automatically prevents dangerous mathematical errors such as numbers becoming too large or too small.

User privacy is protected through multiple mechanisms. Pseudonymity ensures that users do not require any central system and can simply use their wallet to log in. Data minimization ensures that no personal details are stored on the blockchain. Pattern obfuscation spreads transactions at random times during testing to make it difficult to track user behavior.

This detailed analysis shows that the BCFT system is secure for large organizations while also supporting reliable tracking for personal use.

V. CONCLUSION & FUTURE WORK

A. Research Contributions

This work shows a blockchain-based system for tracking emissions and makes three key improvements: **Decentralized Accountability:** A system that uses smart contracts to permanently record emissions from transportation in both personal and business travel. **Emission Insights:** Time-based emissions analysis for daily, weekly, monthly, and annual periods, with separate details for each category, to help create targeted plans to reduce emissions. **Behavioral Reward System Foundation:** Infrastructure that supports future systems where tokens are used to reward users for actions that help achieve the goals.

B. Sustainability Impact

By converting travel data into auditable CO₂ metrics, the BCFT system: Empowers individuals to quantify and contextualize their environmental impact; provides corporations with tamper-proof data suitable for ESG reporting; lowers verification costs for carbon credit markets by an estimated 40–60%; and supports UN Sustainable Development Goal 13 (Climate Action) through transparent emission baselining.

C. Future Enhancements

AI Based Predictive Analytics: The future use of machine learning models includes predicting future emission on the basis of past behavior patterns; suggesting the best transportation routes and methods having low carbon emission; and identifying unusual increases in emission by validating the patterns that predict future trends.

Carbon Market Integration: Hybrid smart contracts will be developed so that when emissions fall below the personal limit, carbon credits are created automatically, giving people the freedom to trade carbon credits directly with each other through blockchain-based financial systems.

Privacy-Preserving Tracking: Advanced cryptographic methods like zk-SNARKs will allow people to show that they meet emission rules without sharing specific travel details, while keeping the process open for everyone to review but ensuring that personal information and time-related data remain safe.

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